

Time: 4hrs	School Of Engineering, CUSAT B.Tech Degree III Semester Internal Examination Mechanical Engineering B-Batch 19-205-0306 Machine Drawing	Marks:60
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Follow 1st angle projection

		CO	BL	Mark
1.a	Draw I.S.O metric V-thread profiles both internal and external, in section showing at least three threads. Indicate all standard proportions in the drawing. Take, pitch as 30mm.	CO2	L2	5
1.b	Draw Lewis type foundation bolt of bolt diameter 24mm and indicate all the dimensions in the drawing	CO2	L2	10
Or				
2	Draw the Front view, Top view and Side view of a hexagonal headed bolt, nut and washer assembly of size M 20. The length of the bolt is 80 mm and the threaded length is 48 mm.	CO1, CO3	L3	15
3	Draw the top half sectional elevation and left side view of flanged coupling shown in Figure-1	CO1, CO3	L3	20
Or				
4	A Plummer block shown in Figure-2. Draw i) Front view with Right half in section and ii) Full Left Side view of the Plummer block assembly.	CO1, CO3	L3	20
5	Disassembled isometric view of a stuffing box is shown in Figure-3. Draw the following views with all parts assembled together i. Right half sectional elevation ii. Plan	CO1, CO4	L4	25
Or				
6	Details of Lathe Tail stock are shown in Figure-4. Assemble the parts and draw the elevation.	CO1, CO4	L4	25

Course Outcomes

On completion of this course the student will be able to: 1. Convert a pictorial view in to orthographic view

2. Draw the different views of threaded nuts and bolts 3. Draw the orthographic views of cotter joints, pipe joints, couplings and bearings 4. Draw the assembly drawing of machine parts

CO/BL	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
% of Total Marks	83.3%	0.83%	45.8%	41.6%			16.6%	41.6%	83.3%		

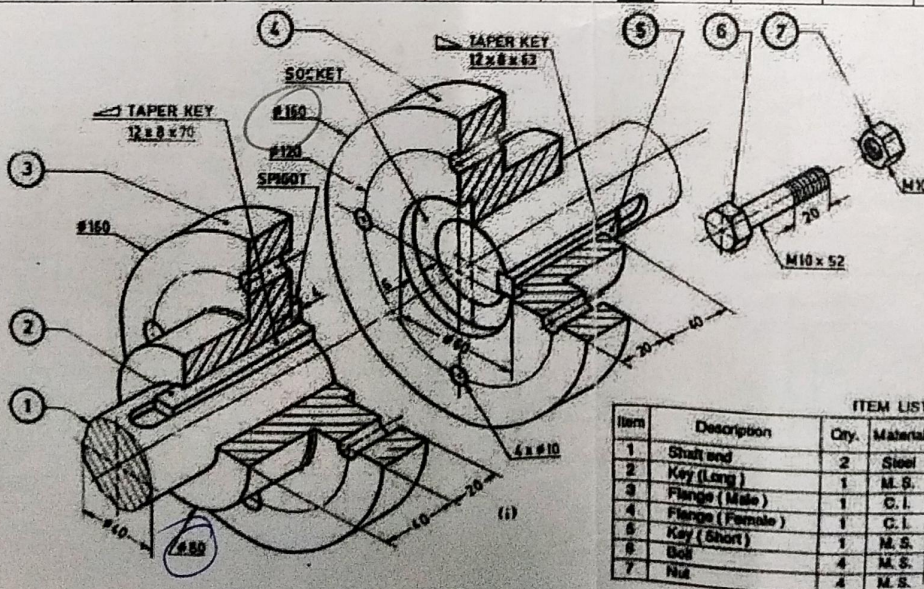


Figure-1

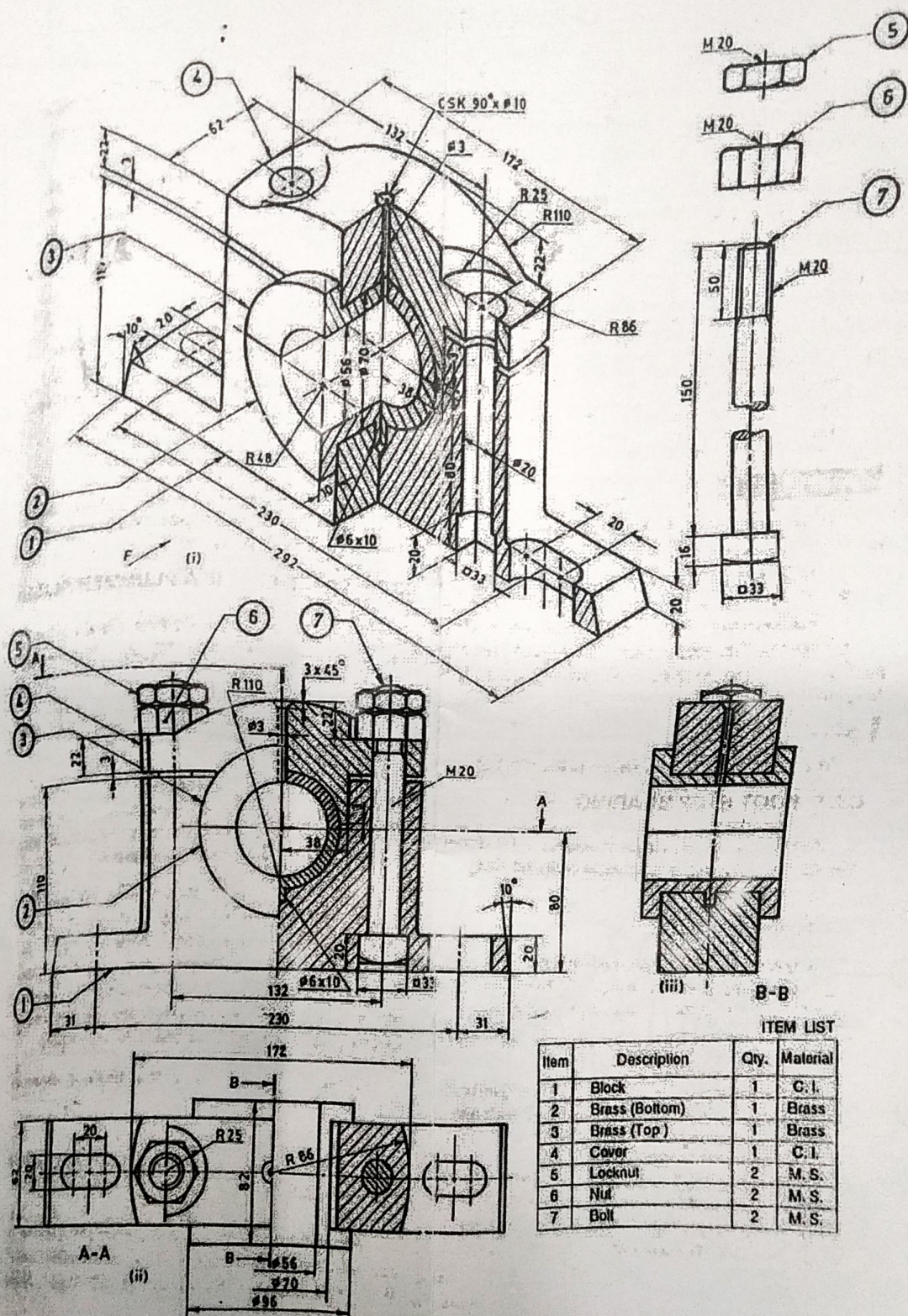
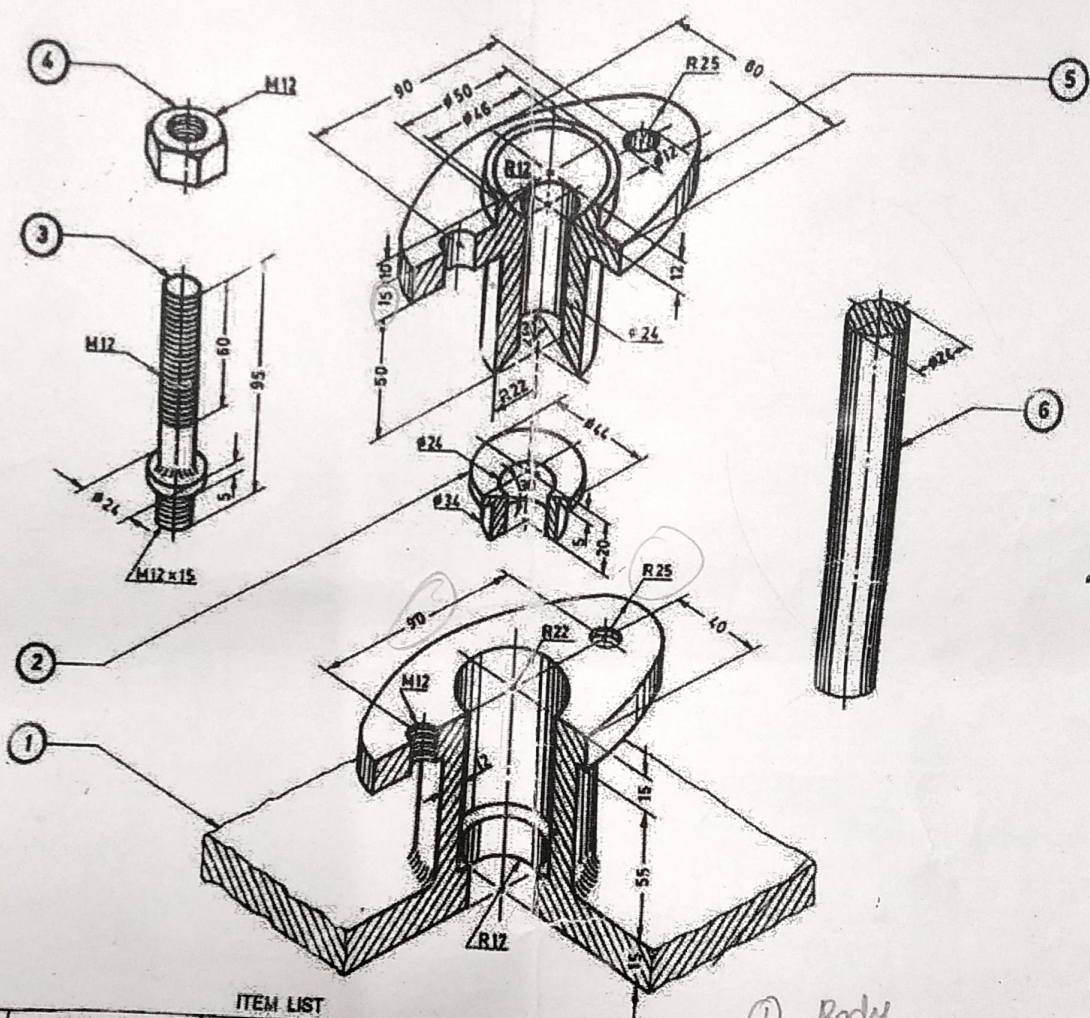


Fig. 2

h=50

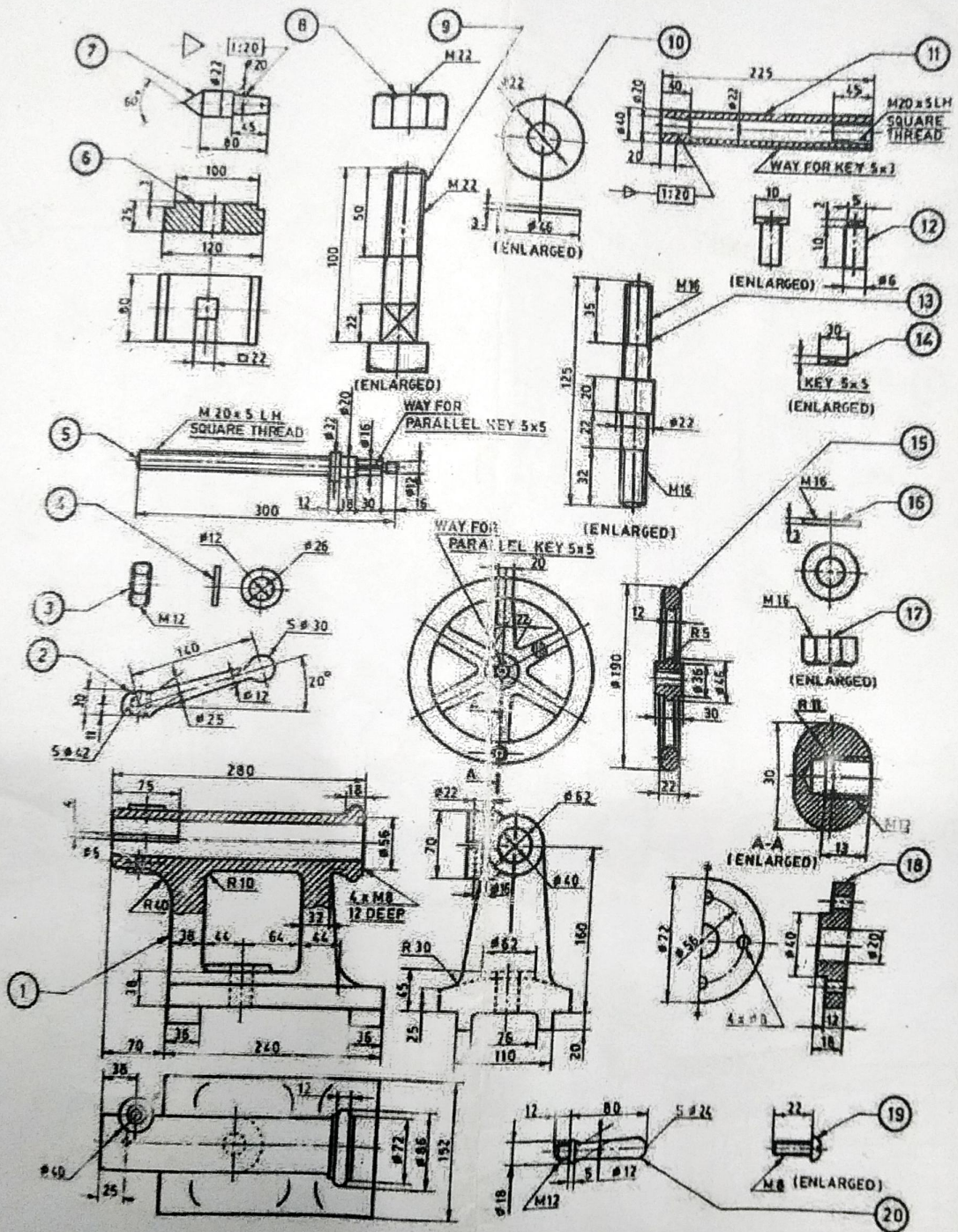


ITEM LIST

Item	Description	Qty.	Material
1	Body	1	C.I.
2	Gland bush	1	Brass
3	Stud	2	M.S.
4	Nut	2	M.S.
5	Gland	1	C.I.
6	Piston rod	1	Steel

Figure-3

- ① Body
- ② Piston Rod
- ③ Gland bush
- ④ Gland



ITEM LIST

Item	Description	Qty.	Material
1	Body	1	C.I.
2	Handle (Lock)	1	C.I.
3	Nut (M12)	1	M.S.
4	Washer (M12)	1	M.S.
5	Socket screw	1	M.S.
6	Coupling plate	1	Steel
7	Door cover	1	Steel
8	Nut (M22)	1	M.S.
9	Co. lock (M22)	1	M.S.
10	Washer (M22)	1	M.S.

Item	Description	Qty.	Material
11	Barrel	1	C.I.
12	Key (Barrel)	1	M.S.
13	Stud	1	M.S.
14	Key (Wheel)	1	M.S.
15	Hand Wheel	1	M.S.
16	Worm (M18)	1	C.I.
17	Mid (M18)	1	M.S.
18	Flange	1	M.S.
19	Socket (M18)	4	C.I.
20	Handle (Wheel)	1	M.S.

Figure-4